

A Push-pull Mechanism Between Prrt2 and B 4 Differentially Regulates Membrane Exposure and Biophysical Properties of Nav1.2 Sodium Channels

Pierluigi Valente

Università degli Studi di Genova: Università degli Studi di Genova

Antonella Marte

Università degli Studi di Genova: Università degli Studi di Genova

Francesca Franchi

Istituto Italiano di Tecnologia

Bruno Sterlini

Università degli Studi di Genova

Silvia Casagrande

Università degli Studi di Genova

Anna Corradi

Università degli Studi di Genova

Pietro Baldelli

Università degli Studi di Genova

Fabio Benfenati (✉ fabio.benfenati@iit.it)

Istituto Italiano di Tecnologia <https://orcid.org/0000-0002-0653-8368>

Research Article

Keywords: Proline-Rich Transmembrane protein 2, intrinsic excitability, transient Na⁺ current, persistent Na⁺ current, resurgent Na⁺ current.

Posted Date: May 6th, 2022

DOI: <https://doi.org/10.21203/rs.3.rs-1585258/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Abstract

Proline-Rich Transmembrane protein 2 (PRRT2) is a neuron-specific protein implicated in the control of neurotransmitter release and neural network stability. Accordingly, PRRT2 loss-of-function mutations associate with pleiotropic paroxysmal neurological disorders, including paroxysmal kinesigenic dyskinesia, episodic ataxia, benign familial infantile seizures, hemiplegic migraine. PRRT2 is a negative modulator of the membrane exposure and biophysical properties of Na⁺ channels Na_v1.2/Na_v1.6 predominantly expressed in brain glutamatergic neurons. Na_v channels form complexes with β-subunits that facilitate the membrane targeting and the activation of the α-subunits. The opposite effects of PRRT2 and β-subunits on Na_v channels raises the question of whether PRRT2 and β-subunits interact or compete for common binding sites on the α-subunit, generating Na⁺ channel complexes with distinct functional properties. Using a heterologous expression system, we have observed that β-subunits and PRRT2 do not interact with each other and act as independent non-competitive modulators of Na_v1.2 channel trafficking and biophysical properties. PRRT2 antagonizes the β4-induced increase in expression and functional activation of the transient and persistent Na_v1.2 currents, without affecting resurgent current. The data indicate that β4-subunit and PRRT2 form a push-pull system that finely tunes the membrane expression and function of Na_v channels and the intrinsic neuronal excitability.

Introduction

Mutations in the Proline-Rich Transmembrane protein 2 (PRRT2) gene cause a broad and heterogeneous spectrum of neurological diseases sharing a paroxysmal nature, such as paroxysmal kinesigenic dyskinesia, episodic ataxia, benign familial infantile seizures, and hemiplegic migraine (1). Nonsense, missense, and frameshift mutations have been reported. The vast majority of these mutations (~ 80%) carry the same frameshift single-nucleotide duplication c.649dupC (1–3) that, via mRNA decay or translation of a non-functional truncated protein, give raise to haploinsufficiency (4–7). The diverse and pleiotropic clinical manifestations of paroxysmal *PRRT2*-linked diseases with no clear genotype-phenotype correlations suggest that the PRRT2 protein has the function of preserving the stability of neuronal networks in which it is expressed (8).

Structurally, PRRT2 is a neuron-specific type-2 integral membrane protein with a large cytosolic N-terminal domain (9), expressed in the axonal and presynaptic domains (10). At the presynaptic level, PRRT2 orchestrates the Ca²⁺-sensitivity of synaptic vesicle exocytosis and by interacting with the SNARE complex, synaptotagmins 1/2, P/Q-type voltage-gated Ca²⁺ channels (Ca_v) and the actin cytoskeleton (10–14). At this level, PRRT2 enhances the probability of release to single stimuli and favors synaptic depression during sustained activity (10, 14, 15).

In addition to regulation of synaptic transmission, an important function of PRRT2 is the control of intrinsic excitability through specific interactions with both voltage-gated Na⁺ (Na_v) 1.2/1.6 channels and the α1/α3-subunits of Na⁺/K⁺ ATPase (15–18). The diffuse neuronal network hyperexcitability observed

in PRRT2 knockout (KO) mice recapitulates the pathological hallmarks of PRRT2-linked disorders (8, 19–21). It has been found that PRRT2 exerts an inhibitory constraint on membrane targeting and exposure of Na_v1.2/Na_v1.6, the two main Na⁺ channel subtypes expressed in excitatory neurons and modulates their biophysical properties (15–17). Such observation represents the molecular basis of the reported efficacy of low doses of the Na_v-blocker carbamazepine in ameliorating the PRRT2-linked clinical manifestations (4, 22–25).

Na_v channels are embedded in a multicomponent membrane signaling complex that involves various integral membrane proteins (26) in which β-subunits are the prominent members forming Na_v heteromeric complexes. These specific supramolecular complexes are composed of a single pore-forming α-subunit, a non-covalently associated β1/β3-subunit, and a covalently linked β2/β4-subunit via a single disulfide bond (27). β-subunits present a large extracellular Ig-like domain, a single multifunctional transmembrane segment, and a short cytoplasmic tail allowing them to positively modulate the gating properties, membrane targeting and expression levels of the α-subunits (28). The recently characterized negative modulatory effect of PRRT2 on Na_v1.2/Na_v1.6 expressed in the absence of β-subunits, raises the question of whether PRRT2 and β-subunits interact or compete for the binding and modulation of Na_v targeting and biophysics, generating Na⁺ channel complexes with distinct functional properties.

Using a heterologous expression system, we have observed that β-subunits do not interact directly with PRRT2. Moreover, the β4-subunit and PRRT2 operate as non-competitive modulators of Na_v1.2 channel trafficking, membrane expression and biophysical properties. PRRT2 antagonizes both the β4-induced increases in expression of Na_v1.2 subunits and the functional activation of the transient and persistent Na_v1.2 currents, without affecting resurgent current. The data indicate that β4-subunit and PRRT2 form a push-pull system that finely tunes the membrane expression and function of Na⁺ channels and that such dual modulation is perturbed by PRRT2 loss-of-function mutation, resulting in hyperactivity of Na⁺ channels and network hyperexcitability.

Materials And Methods

Cell culture and transfection. HEK293 cells were grown in DMEM-F12 supplemented with 0.5 mM pyruvate 10% fetal bovine serum, 1% L-glutamine, 50 U/ml penicillin, 50 µg/ml streptomycin and maintained at 37°C in 5% CO₂. For HEK293 cells stably expressing human Na_v1.2 (SCN2A), 500 µg/ml G418 was added to the medium to select of Na_v channel expressing cells. All reagents were purchased from ThermoFisher Scientific. For testing β1-β4-subunit/Na_v1.2 interactions, Na_v1.2 expressing HEK293 cells were transfected with either FLAG-tagged β-subunit (*SCN1B* [NM_001037], *SCN2B* [NM_004588], *SCN3B* [NM_018400], *SCN4B* [NM_174934]; Origene) or FLAG-tagged bacterial alkaline phosphatase (BAP; Sigma-Aldrich). To evaluate PRRT2 and β1-β4-subunit interactions, HEK293 cells were transfected with either HA-tagged PRRT2 (Rossi et al., 2016) or the unrelated bacterial alkaline phosphatase (BAP) as

a control (16). For competition assays, Na_v1.2-expressing HEK293 cells were transfected with HA-tagged PRRT2 and incubated with FLAG-tagged β 4-subunit expressed in wild type cells. Alternatively, HA-tagged PRRT2 and FLAG-tagged β 4-subunit were co-expressed in Na_v1.2-expressing HEK293 cells. For biotinylation assays and electrophysiological experiments, Na_v1.2-expressing HEK293 cells were transfected with empty vector (pkH3; Addgene), HA-tagged PRRT2 and/or FLAG-tagged β 4-subunit. All transfections were conducted with Lipofectamine 2000 (Invitrogen) according to the manufacturer protocol 48 h before the experiments.

SDS-PAGE and Western blotting. SDS-PAGE was performed according to Laemmli (29). Samples heated to 50°C for 5 min without boiling were run on SDS-PAGE polyacrylamide gels and blotted onto nitrocellulose membranes (Whatman). Blotted membranes were blocked for 1 h in 5% milk in Tris-buffered saline (10 mM Tris, 150 mM NaCl, pH 8.0) plus 0.1% Triton X-100 and incubated overnight at 4°C with the appropriate primary antibody. Membranes were washed and incubated at room temperature for 1 h with peroxidase-conjugated anti-mouse (1:3000; BioRad, Hercules, CA) or anti-rabbit (1:3000; BioRad, Hercules, CA) antibodies. Bands were revealed with the ECL chemiluminescence detection system (ThermoFisher Scientific). Immunoblots were quantified by densitometric analysis of the fluorograms (Quantity One software; Bio-Rad, Hercules, CA) obtained in the linear range of the emulsion response.

Pull down assays. Wild type HEK293 cells or Na_v1.2-expressing HEK293 cells were transfected as previously described. After 48 h, cells were harvested in lysis buffer (150 mM NaCl, 50 mM Tris, 1 mM EDTA and 1% Triton X-100 supplemented with protease inhibitor cocktail) and centrifuged at 10,000 x g for 10 min at 4°C. Kept an aliquot for the input sample, the supernatant was incubated with 50 μ L of either anti-FLAG® M2 Affinity Gel or monoclonal anti-HA-agarose affinity beads (Sigma-Aldrich) at 4°C for 2 h. After extensive washes in lysis buffer and detergent-free lysis buffer, samples were resolved by Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and subjected to western blotting with anti-panNa_v (1:300; Sigma-Aldrich), anti-FLAG (1:2,000; Sigma-Aldrich) or anti-HA (1:1,000; ThermoFisher Scientific) specific antibodies.

Surface biotinylation assays. Forty-eight hours after transfection, Na_v1.2-expressing HEK293 cells were incubated with 1 mg/ml of EZ-Link™ Sulfo-NHS-SS-Biotin (ThermoFisher Scientific) in cold phosphate-buffered saline (PBS) for 35 min at 4°C, with constant mixing. Free biotin was quenched, twice with 100 mM Tris pH 8, and once with cold PBS to remove the excess of biotin. Cells were then lysed in lysis buffer (150 mM NaCl, 50 mM Tris, 1 mM EDTA and 1% Triton X-100) supplemented with protease inhibitor cocktail (Cell Signaling). Total cell lysates were centrifuged at 10,000 x g at 4°C for 10 min. Kept an aliquot for the input sample, the supernatant fraction was incubated with 150 μ L of NeutrAvidin-conjugated agarose beads (ThermoFisher Scientific) at 4°C for 3 h. After extensive washes of the beads, samples were eluted, resolved by SDS-PAGE and subjected to western blotting with anti-panNa_v (1:300; Sigma-Aldrich), anti-Na/K ATPase pump 1 (1:1,000; Merck), anti-FLAG (1:2,000; Sigma Aldrich) or HA (1:1,000; ThermoFisher Scientific) antibodies.

Electrophysiological recordings and analysis. Whole-cell voltage-clamp experiments were performed on HEK293 stably expressing Na_v1.2 α -subunit transiently transfected with cDNA (2 μ g) of empty vector pKH3 (MOCK), PRRT2-HA, β 4-subunit or PRRT2-HA + β 4-subunit. Transfection was done with Lipofectamine 2000, following the manufacturer's recommendations. Transfected cells were enzymatically dissociated, re-plated at low density about 24 h post-transfection. Recordings were performed 24 h after replating; transfected cells were identified by fluorescence of co-transfected Tomato protein reporter (Clontech). Recording solutions were previously described (16). Briefly: the standard external solution contained (in mM): 140 NaCl, 3 KCl, 1 MgCl₂, 1 CaCl₂, 10 HEPES, 10 Mannitol (pH 7.3 with NaOH); the standard internal solution contained (in mM): 140 CsCl, 10 NaCl, 2 EGTA, 10 HEPES (pH 7.3 with CsOH). To induce resurgent currents, 200 μ M Na_v β 4 peptide (KKLITFILKKTREK-OH; Proteogenix) corresponding the COOH-terminal tail of full-length β 4-subunit, was included in the pipette solution. Patch pipettes, prepared from thin-borosilicate glass (Hilgenberg) were pulled and fire-polished to a final resistance of 2–3 M Ω when filled with standard internal solution. Electrophysiological experiments were done using an EPC-10 amplifier (HEKA Elektronik). Whole-cell voltage-clamp recordings of Na_v currents were acquired at 20 kHz and low-pass filtered at 4 kHz. Recordings with leak currents > 200 pA or series resistance > 10 M Ω were discarded. Data acquisition was performed using the PatchMaster program (HEKA Elektronik GmbH). Series resistance was compensated 80% (2 μ s response time) and the compensation was readjusted before each stimulation. The membrane potentials in whole-cell recordings were uncorrected for Donnan liquid junction potentials of $\sim$ 9 mV (30). All experiments were performed at room temperature (22–24°C). Whole-cell family currents of fast inactivating Na_v channels were evoked by 5 mV steps depolarization (100 ms in duration) from -80 to 65 mV and cells were held at a holding potential (V_h) of -120 mV. Steady-state inactivation curves were constructed by recording the peak current amplitude evoked by 20-ms test pulses to -10 mV after 500-ms pre-pulses to potentials over the range of -130 to 10 mV ($V_h = -120$ mV). The conductance-voltage (G-V) curves were obtained by converting the maximal current values, evoked with a voltage step protocol in each cell, to conductance according to the extended Ohm's law: $G = I_{Na}/(V - E_{Na})$, where I_{Na} is the peak Na⁺ current measured at potential V , and E_{Na} is the calculated Nernst equilibrium potential. G-V curves were normalized and fitted with the Boltzmann function $G/G_{max} = 1/(1 + \exp[(V - V_{1/2})/k])$, where G is the conductance, G_{max} is the maximal conductance, $V_{1/2}$ is the half-maximal voltage of activation, and k is the slope factor. Inactivation curves were fitted with the Boltzmann equation in the following form: $1/[1 + \exp(V_{1/2} - V)/k]$. Time-dependent rate of recovery from inactivation was calculated by pre-pulsing the cell with a 20-ms step to -20 mV to inactivate the channels and then bringing back the potential to -100 mV for increasing recovery durations (0.5, 1, 2, 4, 8, 32, 64, 128, 148 ms) before the test pulse of -20 mV ($V_h = -120$ mV). Time constants for recovery from inactivation were obtained by fitting data from each cell to a first order exponential function and averaging time constants across cells. Persistent currents were evoked from $V_h = -120$ mV by 5-mV depolarization steps of 600 ms from -60 to 50 mV. Resurgent currents were evoked with depolarization steps from V_h of -120 mV to 30 mV (20 ms) to open channels, allowing them to undergo open-channel block, and subsequently repolarizing to potentials ranging from -50 to 20 mV (60 ms) to allow the blocker to unbind. Persistent currents were measured in the last 60 ms of a 600 ms depolarizing step

pulse. Resurgent currents were measured after 2.5 ms into the repolarization step to bypass fast tail currents. The percentage of persistent/resurgent currents were calculated by dividing the current amplitude by the peak transient current in each recorded cell. To minimize space-clamp problems, we selected only isolated cells with a soma diameter of about $< 30 \mu\text{m}$ for recordings. Membrane capacitance artefacts and leakage currents were eliminated by P/N leak subtraction procedure.

Statistical analysis. All data points are presented as mean $\pm$ standard error of the mean (sem) for number of cells or number of independent experiments (n), as detailed in the figure legends. Normal distribution of data was assessed using the D'Agostino-Pearson's normality test. The F test was used to compare variance between two sample groups. To compare two normally distributed sample groups, the Student's unpaired *t*-test was used. To compare two sample groups that were not normally distributed, the non-parametric Mann-Whitney's *U*-test was used. To compare more than two normally distributed sample groups, we used one-way ANOVA, followed by either the Bonferroni's or the Fischer's test, was used. In cases in which data were not normally distributed, non-parametric one-way ANOVA (Kruskal-Wallis' test) was used, followed by the Dunn's multiple comparison test. Alpha levels for all tests were 0.05% (95% CIs). Statistical analysis was conducted using OriginPro-8 (OriginLab) and Prism (GraphPad Software) software.

Results

PRRT2 and the Na_v β 4-subunit do not interact with each other or compete for binding to Na_v1.2

Na_v β -subunits are known to positively modulate the Na⁺ channel membrane exposure and biophysical properties. Thus, it was of interest to investigate whether the inhibitory constraint of PRRT2 on Na⁺ channel activity was attributable to PRRT2-induced sequestration of β -subunits or interference with the binding of β -subunits to the Na_v α -subunit. To this aim, we performed affinity binding assays to assess the interactions between Na_v1.2, β 4-subunits and PRRT2. Among β -subunits, β 4 was chosen for its high expression in the cerebellum (28), which is the main brain region responsible for the phenotype in PRRT2 KO mice and where PRRT2 reaches the highest expression under physiological conditions (8, 19).

Moreover, of the four β -subunits, β 4 is the only subunit that enables the resurgent Na⁺ current responsible for high-frequency firing in neurons (31). HA-tagged PRRT2 and β -subunits were transiently expressed in HEK293 cell clones stably transfected with human Na_v1.2 and the PRRT2/Na_v1.2 α/β complexes were pulled down with anti-HA beads and identified by western blotting with anti-Na_v α - and β -subunit antibodies.

We asked whether PRRT2 can bind directly to the cytosolic or transmembrane domains of β -subunits, thus decreasing their availability for the Na_v1.2 α/β complex. Affinity binding assays performed in wild type HEK293 cells expressing HA-tagged PRRT2 and either the negative control BAP or the various β -subunits revealed that PRRT2, pulled down by anti-HA beads, did not significantly associate with either β -subunit detected by western blotting with β 1/ β 4-subunit specific antibodies (Fig. 1A). Next, using the

Na_v1.2 α -subunit as a hub, we investigated whether PRRT2 and the β 4-subunit can bind to the Na_v1.2 α -subunit independently at distinct sites or they compete for a single association site. When both HA-tagged PRRT2 and β 4-subunit were expressed in HEK293 cell clones stably transfected with human Na_v1.2, pull down of PRRT2 with anti-HA beads resulted in the co-precipitation of similar amounts of Na_v1.2 α -subunit together with detectable amounts of β 4-subunit (Fig. 1B). To assay for a direct competition of PRRT2 and β 4-subunit for a shared binding site on the α -subunit, HA-tagged PRRT2 or BAP was expressed in HEK293 cells stably transfected with human Na_v1.2 and the HA-PRRT2/Na_v1.2 immunoprecipitated complex was challenged in the absence or presence of an excess of β 4-subunit expressed in wild type HEK293 cells. Also under these conditions, the presence of an excess β 4-subunit did not inhibit the binding of PRRT2 to Na_v1.2, as the amount of α -subunit present in the immunocomplexes was not affected in the presence or absence of β 4-subunit, as shown by western blotting with anti-Na_v antibodies (Fig. 1C).

Taken together, the data indicate that PRRT2 and the Na_v β 4-subunit do not interact with each other and bind independently to the Na_v1.2 α -subunit.

PRRT2 and Na_v β 4-subunit have opposite actions of the trafficking of the Na_v1.2 α -subunit to the plasma membrane

It is well known that the Na_v β -subunits enhance the trafficking of the α -subunits to the plasma membrane, resulting in a larger population of active channels exposed to the extracellular *milieu* (28). On the other hand, we recently shown that PRRT2 plays the opposite action of inhibiting the membrane exposure of Na_v1.2/Na_v1.6 that causes a decrease in Na⁺ current (16). Thus, we investigated the effect of the simultaneous presence of both α -subunit modulators on the surface exposure of the channel. To this aim, we performed surface biotinylation of HEK293 cell clones stably expressing the human Na_v1.2 α -subunit that had been transiently transfected with HA-tagged PRRT2, β 4-subunit or both and quantified the amount of surface-labelled Na_v1.2 α -subunits. The results show that, while PRRT2 and β 4-subunit have the reported opposite actions on the exposure of Na_v1.2 channels on the plasma membrane, the combined expression of the two modulators was not significantly different from the control sample, indicating that the opposite actions of PRRT2 and β 4-subunit on Na_v1.2 channel trafficking occur independently and are simply additive (Fig. 2).

PRRT2 expression counteracts the increase in Na_v1.2 current density induced by β 4-subunit

To test the influence of PRRT2 on the properties of Na⁺ currents mediated by the physiological complex Na_v α -subunit- β 4-subunit, macroscopic whole-cell currents were recorded from HEK293 cells stably expressing Na_v1.2 which had been sequentially transfected with empty vector (pKH3; MOCK), PRRT2, β 4-subunit or with PRRT2 + β 4-subunit (Fig. 3A). Families of transient Na⁺ currents were elicited by applying

100-ms depolarizing steps ranging from -80 mV to 65 mV from a holding potential of -120 mV (*inset*). Current density (J)/voltage (V) curves were built for all experimental conditions by normalizing the peak current at each voltage by the cell capacitance (Fig. 3B). When compared to control cells, PRRT2-expressing cells showed the previously reported reduction of the transient Na^+ current across a wide voltage range and in the absence of voltage shifts in the peak Na^+ current (16; Fig. 3B,C). As expected, expression of the $\beta 4$ -subunit had a positive modulatory effect on Na^+ current density with a two-fold increase of the $\text{Na}_v 1.2$ current density, as compared to MOCK-transfected cells with a shift of the peak current towards more negative voltages (Fig. 3B,C). The observation that the properties of $\text{Na}_v 1.2$ currents were modified after either $\beta 4$ -subunit or PRRT2 transfection suggests that both these proteins were successfully integrated into the channel signaling complex and exerted their modulatory action.

Interestingly, when PRRT2 was co-expressed with the $\beta 4$ -subunit it neutralized the positive modulation of the β -subunit on amplitude of the Na^+ current density, while the $\beta 4$ -subunit induced leftshift of the J/V curve was preserved (Fig. 3B,C). The data confirm that PRRT2 and $\beta 4$ -subunit affect membrane targeting and exposure of the $\text{Na}_v 1.2$ α -subunit in opposite directions, indicating that they do that via independent mechanisms of action with resulting additive effects.

PRRT2 and $\beta 4$ -subunit differentially affect the kinetics of activation and inactivation of $\text{Na}_v 1.2$ channels

To dissect the effects of PRRT2 on the biophysical properties of $\text{Na}_v 1.2$ α -subunits in the presence of the $\beta 4$ -subunit, we next examined the voltage-dependence of channel activation and steady-state inactivation. In a previous study (16), we observed that PRRT2 did not affect the $\text{Na}_v 1.2$ activation curves, while it favored channel inactivation at more negative potentials (Fig. 4A,B). On the other hand, the $\beta 4$ -subunit is known to cause a leftward shift of the activation curve of $\text{Na}_v 1.2$, without affecting the voltage-dependence of inactivation (27).

The specific modulation of the $\text{Na}_v 1.2$ biophysical properties by PRRT2 and the $\beta 4$ -subunit were fully preserved when the two auxiliary proteins were co-expressed. Under this condition, the activation curves displayed the leftward shift as signature of the $\beta 4$ -subunit, while the steady-state inactivation curves were shifted at more negative potentials by the concomitant action of PRRT2 (Fig. 4A,B). Notably, $\beta 4$ exerted its action on the activation irrespective of the absence or presence of PRRT2 and PRRT2 exerted its action on the inactivation irrespective of the absence or presence of $\beta 4$.

The quantitation of the main biophysical parameters obtained from the Boltzmann fitting of individual activation and inactivation curves confirmed that the voltage of half-maximal activation ($V_{0.5}$), slope and maximum conductance of activation (G_{max}) were significantly changed in the presence of the $\beta 4$ -subunit and not affected by PRRT2 (Fig. 4C), while the voltage of half-maximal inactivation ($V_{0.5}$) was decreased by PRRT2 and not affected by of the $\beta 4$ -subunit, that only induced a slight decrease of the slope of inactivation (Fig. 4D).

PRRT2, but not the $\beta 4$ -subunit, modulates the $\text{Na}_v 1.2$ recovery kinetics from inactivation.

We measured channel recovery from inactivation using a voltage command protocol in which we evaluated the peak current by: (i) measuring current evoked by a first depolarizing step to -20 mV; (ii) allowing current to recover from inactivation at -100 mV for progressively increasing time intervals; (iii) measuring current recovery with a second test pulse to -20 mV (Fig. 5A, *inset*). When compared to MOCK-transfected cells, the extent of channel recovery was significantly decreased by PRRT2, while it was unaffected by the $\beta 4$ -subunit. When both auxiliary proteins were co-expressed, the PRRT2-induced decrease of the recovery plateau was only slightly attenuated, but still significant with respect to MOCK-transfected cells, while the time constant of recovery was not significantly modified under the various experimental conditions (Fig. 5B,C).

Moreover, the $\beta 4$ -subunit expression reduces the use-dependent inhibition of $\text{Na}_v 1.2$ by PRRT2 as shown by the ratio between the current evoked by the last and first step of the protocol used to study the recovery from fast channel inactivation (Fig. 5D). All these results indicate that PRRT2 and $\beta 4$ -subunit operate together into the $\text{Na}_v 1.2$ channel signaling complex by producing discrete and distinct modulations of the channel biophysics.

PRRT2 and $\beta 4$ -subunit independently modulate the $\text{Na}_v 1.2$ persistent and resurgent currents

Increases in both resurgent and persistent currents through Na_v channels are associated with neuronal hyperexcitability and increase in firing rates (32, 33). Thus, we analyzed the effects of the combined expression of PRRT2 and $\beta 4$ -subunit on these currents evoked in $\text{Na}_v 1.2$ -expressing HEK293 cells using specific protocols. Since the persistent Na^+ current is a non-inactivating (or very slowly-inactivating) current, we measured it over the last 60 ms of the 600 ms incremental conditioning steps for each condition tested (Fig. 6A, *inset*). Due to the variability of current density across cells, the persistent current amplitude was normalized to the peak amplitude of the transient current for each cell, and the percentage of persistent current was plotted *versus* the applied voltage. We found that when PRRT2 and $\beta 4$ -subunit were individually expressed, the percent amplitude of the $\text{Na}_v 1.2$ persistent current was decreased by PRRT2 and increased by the $\beta 4$ -subunit, when compared to mock-transfected cells. When both auxiliary proteins were co-expressed, the amount of persistent current was similar that recorded in mock-transfected cells (Fig. 6B,C). These findings, which closely resemble the behavior of the macroscopic transient Na^+ current (see Fig. 3), demonstrate that PRRT2 and $\beta 4$ -subunit modulate independently the $\text{Na}_v 1.2$ persistent current when expressed in the same channel complex and that their effects are purely additive.

The resurgent Na^+ current is caused by the influx of Na^+ ions through Na_v channels during repolarization (34). To elicit that, we applied an initial depolarizing step from -120 to 60 mV, followed by subsequent incremental repolarizing steps from -50 mV to 20 mV (Fig. 6D, *inset*). Because the expression of the full-

length $\beta 4$ -subunit is not sufficient to produce resurgent currents in $\text{Na}_v 1.2$ -expressing HEK293 cells, we included in the pipette solution a $\beta 4$ peptide ($\beta 4$ -ptd) derived from the COOH-terminal of the $\beta 4$ -subunit (see Materials and Methods), that is known to induce resurgent currents (28, 35, 36).

Then we recorded Na^+ resurgent currents in $\text{Na}_v 1.2$ -expressing HEK293 cells which had been transfected either with the empty vector or with PRRT2 cDNA in the absence or presence of the $\beta 4$ -peptide in the pipette solution (MOCK and MOCK + $\beta 4$ -ptd, respectively) (Fig. 6D). Also in this case, to reduce the variability in current density across recorded cells, the resurgent current at each repolarizing voltage step was normalized to the peak amplitude of the transient current recorded in the same cell under the same experimental condition. While, as expected, the presence of $\beta 4$ -peptide greatly increased the resurgent current in a wide range of voltages, PRRT2 was totally ineffective in modulating the resurgent current in MOCK-transfected cells in both presence and absence of the $\beta 4$ -peptide (Fig. 6E,F). Hence, these data it appears that the expression and modulation of the $\text{Na}_v 1.2$ persistent current depends exclusively on the presence of the $\beta 4$ -peptide and is independent of PRRT2.

Discussion

PRRT2 and β subunits are both Na_v modulatory proteins

The neuron specific PRRT2 is an important modulator of presynaptic functions and intrinsic excitability. Neuronal circuits lacking PRRT2 become hyperexcitable (37). Accordingly, patients bearing loss-of-function mutations in the *PRRT2* gene or PRRT2 KO mice are affected by paroxysmal manifestations, whose pleiotropism ranges from paroxysmal kinesigenic dyskinesia to epilepsy and migraine. Based on previous studies, these phenotypes result from a mixed synaptopathy/channelopathy (3, 38). Indeed, we demonstrated that PRRT2 acts, in murine and human neurons, as a negative modulator of 1.2/1.6 Na_v channels, without affecting the 1.1 subtype that is essential for the excitability of inhibitory neurons (16). These findings well correlate with the therapeutic efficacy of Na_v channel inhibitors in ameliorating the clinical phenotype of PRRT2-linked disorders (22, 25, 36). PRRT2 decreases the membrane targeting and exposure of active Na_v channels on the plasma membrane and, in addition, it modulates their biophysical properties by shifting to the left the inactivation curve and decreasing channel recovery from inactivation (16). PRRT2 can thus be envisaged as a novel inhibitory auxiliary subunit of the 1.2/1.6 Na_v α -subunits.

The best-known auxiliary subunits modulating both membrane exposure and kinetics of the pore-forming Na_v α -subunit are the β subunits ($\beta 1$ – $\beta 4$) that, similarly to PRRT2, are single membrane spanning domain proteins. Among β -subunits, $\beta 4$ was chosen for being highly expressed in the cerebellum (28), the main brain region responsible for the PRRT2 loss-of-function phenotype (8, 19) and the only subunit that enables the resurgent Na^+ current (31). When expressed in heterologous systems, the $\beta 4$ subunit increases the Na^+ current density by enhancing the membrane exposure of Na_v α -subunits, shifts the voltage-dependence of channel activation toward more negative voltages and accelerates the rate of activation (27, 28, 39).

PRRT2 and $\beta 4$ exert independent and distinct modulations of density and biophysical properties of $\text{Na}_v 1.2$

Given that both PRRT2 and β subunits associate with the pore-forming α -subunit but exert opposite effects on its membrane exposure and biophysical properties, we considered the possibility that PRRT2 and β subunits interfere with each other in binding to the α -subunit, or they act independently as auxiliary subunits on the common Na_v target. It has been shown that $\beta 2$ and $\beta 4$ subunits covalently interact with Na_v α -subunit by forming an extracellular disulfide bond (40, 41). While direct competition by PRRT2 at this site it is not possible due to the substantial lack of an extracellular domain (9), other potential α/β interaction sites are possible. In fact, interactions between the transmembrane domain of $\text{Na}_v 1.4$ α -subunit (42) and modulation of Na_v biophysical properties by the intracellular domains of $\beta 2$ and $\beta 4$ (43, 44) have been reported, suggesting the existence of other potential interaction sites in the transmembrane/intracellular domains of α and β Na_v subunits. Therefore, it was important to check a possible PRRT2/ β competition in these regions of the channel complex.

Looking at the molecular interactions within the $\text{Na}_v 1.2$ supramolecular complex, we found that: (i) both PRRT2 and β subunits bind to $\text{Na}_v 1.2$ channels; (ii) PRRT2 does not interact with any of the β subunits; (iii) β subunits do not compete for PRRT2 binding to $\text{Na}_v 1.2$ channels; (iv) PRRT2 and β subunits have opposite and additive effects on the targeting and membrane exposure of $\text{Na}_v 1.2$ channels. These data indicate that PRRT2 and $\beta 4$ have distinct docking sites and bind independently to $\text{Na}_v 1.2$ α -subunit, without competing for a common site, resulting in a purely additive effect on the membrane targeting and density of the $\text{Na}_v 1.2$ channel.

The subsequent electrophysiological investigation revealed that PRRT2 and $\beta 4$ have effects that are partly opposite and partly distinct and complementary on current density and biophysics. The effects of the single and combined expression of the two auxiliary proteins are schematically summarized in the radar plot of Fig. 7. The $\beta 4$ subunit increases the transient Na^+ current; PRRT2 only decreases the Na^+ current; the PRRT2/ $\beta 4$ association displays a pure additive effect whereby the Na^+ current remains unchanged but shifted to the left due to the specific $\beta 4$ effect. PRRT2 does not affect the kinetics of activation; $\beta 4$ induces a marked shift of the activation curve to negative voltages and an increase in G_{max} that is maintained unaffected in the presence of PRRT2. On the opposite, $\beta 4$ is ineffective on the kinetics of inactivation and recovery from inactivation; PRRT2 shifts to negative voltages the kinetics of inactivation and decreases the recovery from inactivation, effects that remain unaltered in the presence of $\beta 4$. Overall, these data demonstrate that the two proteins cover distinct, partly overlapping, domains of Na_v channel modulation, but that their effects on the Na^+ channel biophysical properties are purely additive and likely depend on the degree of neuron-specific expression and turnover of the respective auxiliary proteins.

The effects on the persistent Na^+ current reproduced those observed on the transient current, with $\beta 4$ increasing the current amplitude and PRRT2, as recently reported in cerebellar granule cells (17), decreasing it, so that the two opposite effects were totally neutralized in the presence of both proteins. On the other hand, the resurgent Na^+ current generated in the presence of the intracellular $\beta 4$ peptide was specifically increased by $\beta 4$, irrespective of the presence or absence of PRRT2 that was previously shown not to modulate this current in cerebellar granule cells (17).

Push-pull control of Na_v function by PRRT2 and β subunits

These results open the possibility that PRRT2 and Na_v β -subunits are part of a push-pull mechanism that fine tunes the activity of Na^+ channels and their targeting and exposure on the membrane. In addition, the two auxiliary proteins have specific and non-superimposable effects on the channel biophysical properties. The cell density of Na_v channels is the fundamental determinant of intrinsic excitability in neurons. In such a complex system, a dual control of Na^+ channel density, particularly of the PRRT2-sensitive $\text{Na}_v 1.2/1.6$ subtypes responsible for the excitability of excitatory neurons, is the most efficient way to control the “excitability tone” over time. Under this double control, the activity of Na_v α -subunits can be modulated by the relative expression of PRRT2 and β -subunits, thus determining the heterogeneity in intrinsic excitability observed in various neuronal populations.

The majority of the expressed Na_v α -subunits are intracellular, within the endoplasmic reticulum, Golgi apparatus and secretory vesicles (45). Here, α -subunits undergo extensive glycosylation essential for the delivery of the channel to the plasma membrane, its stability and biophysical properties (46, 47). Na_v β -subunits associate with the channel at the intracellular level and promote its trafficking to the plasma membrane (48, 49). Experimental evidence suggests that this requires N-linked glycosylation β -subunits, as well as a positive effect of the β -subunits on glycosylation of the α -subunits that promotes stabilization of the channel at the plasma membrane (50, 51). Moreover, β -subunits also belong to the CAM family of adhesion molecules, and cell adhesion and interactions with the cytoskeleton by β -subunits seem to be critical to the cell surface expression of α -subunit (52, 53).

A question that remains unanswered is how PRRT2 acts in inhibiting the Na_v density on the plasma membrane. In principle, PRRT2 can modulate the membrane surface density of Na_v channels through various potential mechanisms, namely: (i) slowing down exocytosis of intracellular Na_v -containing vesicles; (ii) enhancing the turnover of membrane domains containing Na_v channels; (iii) modulating Na_v channel interactions with the cytoskeleton. Although no direct evidence is as yet available, interactions of PRRT2 with SNARE proteins responsible for membrane fusion at nerve terminals have been reported (5, 10, 11), as well as potential interactions of the proline-rich cytosolic N-terminal domain of PRRT2 with SH3-domain bearing proteins involved in endocytosis, such as endophilin and intersectin (9). Moreover, PRRT2 has been recently shown to modulate the actin-based cytoskeleton and, when expressed in non-neuronal cell lines inhibits cell motility and focal adhesion turnover (12, 13).

Conclusions

The results indicate that PRRT2 can be considered a novel Na_v auxiliary subunit with specificity for the 1.2/1.6 channel subtypes. Although the results were obtained in a heterologous expression system, the previous data indicate that they may hold true also in native neurons. We demonstrate that PRRT2 and β-subunits do not interact, rather they can act in concert by modulating density and properties of voltage-dependent Na⁺ channels. These effects may represent the basis of a homeostatic mechanism that controls the level of intrinsic neuronal excitability in a dynamic fashion based on specific combinations of PRRT2 and β-subunit expression.

Declarations

Funding. The study was supported by research grants from the Compagnia di San Paolo Torino (2015.0546 to FB and 2017.20612 to PB); IRCCS Ospedale Policlinico San Martino (Ricerca Corrente and "5x1000" to AC, PV, PB and FB) and Italian Ministry of University and Research (and 2017-A9MK4R and 2020WMSNBL to FB). The support of Telethon-Italy (Grant GGP19120 to FB) is also acknowledged.

Competing Interests. The authors have no relevant financial or non-financial interests to disclose.

Author Contributions. PV and FF carried out the electrophysiological experiments; AM, BS and SC engineered HEK cells and performed the biochemical experiments; AC and PB supervised the biochemical and electrophysiological experiments, respectively; PV and PB analyzed and interpreted the electrophysiological data; FB and PV planned the experiments and wrote the paper; all authors contributed to the final version of the manuscript.

Acknowledgments. We are grateful to Dr. Marzia Lecchi (Dept. of Biotechnology and Biosciences, University of Milano-Bicocca) for kindly providing stable HEK293 clones expressing Na_v channels.

Data Availability Statement. The datasets generated during and/or analyzed during the current study are available from the corresponding authors on reasonable request.

Ethics Approval. The study was carried out entirely on cell lines and did not involve human or animal subjects. The IRCCS Ospedale Policlinico San Martino Research Ethics Committee has confirmed that no ethical approval is required.

Consent to Participate and Consent to Publish. The study did not involve human subjects. Thus, no informed consent to participate and consent to publish are required.

References

1. Ebrahimi-Fakhari D, Saffari A, Westenberger A, Klein C (2015) The evolving spectrum of PRRT2-associated paroxysmal diseases. *Brain* 138:3476–3495. <https://doi.org/10.1093/brain/awv317>

2. Heron SE, Dibbens LM (2013) Role of PRRT2 in common paroxysmal neurological disorders: a gene with remarkable pleiotropy. *J Med Genet* 50:133–139. <http://dx.doi.org/10.1136/jmedgenet-2012-101406>
3. Valtorta F, Benfenati F, Zara F, Meldolesi J (2016) PRRT2: from paroxysmal disorders to regulation of synaptic function. *Trends Neurosci* 39:668–679. <https://doi.org/10.1016/j.tins.2016.08.005>
4. Chen WJ, Lin Y, Xiong ZQ, Wei W, Ni W, Tan GH, Guo SL, He J, Chen YF, Zhang QJ, Li HF, Lin Y, Murong SX, Xu J, Wang N, Wu ZY (2011) Exome sequencing identifies truncating mutations in PRRT2 that cause paroxysmal kinesigenic dyskinesia. *Nat Genet* 43:1252–1255. doi: 10.1038/ng.1008
5. Lee HY, Huang Y, Bruneau N, Roll P, Roberson ED, Hermann M, Quinn E, Maas J, Edwards R, Ashizawa T, Baykan B, Bhatia K, Bressman S, Bruno MK, Brunt ER, Caraballo R, Echenne B, Fejerman N, Frucht S, Gurnett CA, Hirsch E, Houlden H, Jankovic J, Lee WL, Lynch DR, Mohammed S, Müller U, Nespeca MP, Renner D, Rochette J, Rudolf G, Saiki S, Soong BW, Swoboda KJ, Tucker S, Wood N, Hanna M, Bowcock AM, Szepietowski P, Fu YH, Ptáček LJ (2012) Mutations in the gene PRRT2 cause paroxysmal kinesigenic dyskinesia with infantile convulsions. *Cell Rep* 1:2–12. <https://doi.org/10.1016/j.celrep.2011.11.001>
6. Liu YT, Nian FS, Chou WJ, Tai CY, Kwan SY, Chen C, Kuo PW, Lin PH, Chen CY, Huang CW, Lee YC, Soong BW, Tsai JW (2016) PRRT2 mutations lead to neuronal dysfunction and neurodevelopmental defects. *Oncotarget* 7:39184–39196. <https://doi.org/10.18632/oncotarget.9258>
7. Tsai MH, Nian FS, Hsu MH, Liu W, Liu YT, Liu C, Lin PH, Hwang DY, Chuang YC, Tsai JW (2019) PRRT2 missense mutations cluster near C-terminus and frequently lead to protein mislocalization. *Epilepsia* 60:807–817. <https://doi.org/10.1111/epi.14725>
8. Michetti C, Castroflorio E, Marchionni I, Forte N, Sterlini B, Binda F, Fruscione F, Baldelli P, Valtorta F, Zara F, Corradi A, Benfenati F (2017) The PRRT2 knockout mouse recapitulates the neurological diseases associated with PRRT2 mutations. *Neurobiol* 99:66–83. <https://doi.org/10.1016/j.nbd.2016.12.018>
9. Rossi P, Sterlini B, Castroflorio E, Marte A, Onofri F, Valtorta F, Maragliano L, Corradi A, Benfenati F (2016) A novel topology of proline-rich transmembrane protein 2 (PRRT2): hints for an intracellular function at the synapse. *J Biol Chem* 291:6111–6123. <https://doi.org/10.1074/jbc.M115.683888>
10. Valente P, Castroflorio E, Rossi P, Fadda M, Sterlini B, Cervigni RI, Prestigio C, Giovedì S, Onofri F, Mura E, Guarnieri FC, Marte A, Orlando M, Zara F, Fassio A, Valtorta F, Baldelli P, Corradi A, Benfenati F (2016) PRRT2 is a key component of the Ca²⁺-dependent neurotransmitter release machinery. *Cell Rep* 15:117–131. <https://doi.org/10.1016/j.celrep.2016.03.005>
11. Coleman J, Jouannot O, Ramakrishnan SK, Zanetti MN, Wang J, Salpietro V, Houlden H, Rothman JE, Krishnakumar SS (2018) PRRT2 regulates synaptic fusion by directly modulating SNARE complex assembly. *Cell Rep* 22:820–831. <https://doi.org/10.1016/j.celrep.2017.12.056>
12. Savino E, Cervigni RI, Povoletto M, Stefanetti A, Ferrante D, Valente P, Corradi A, Benfenati F, Guarnieri FC, Valtorta F (2020) Proline-rich transmembrane protein 2 (PRRT2) regulates the actin cytoskeleton during synaptogenesis. *Cell Death Dis* 11:856. <https://doi.org/10.1038/s41419-020-03073-w>

13. Savino E, Guarnieri FC, Tsai JW, Corradi A, Benfenati F, Valtorta F (2021) An emerging role of PRRT2 in regulating growth cone morphology. *Cells* 10:2666. <https://doi.org/10.3390/cells10102666>
14. Ferrante D, Sterlini B, Prestigio C, Marte A, Corradi A, Onofri F, Tortarolo G, Vicidomini G, Petretto A, Muià J, Thalhammer A, Valente P, Cingolani LA, Benfenati F, Baldelli P (2021) PRRT2 modulates presynaptic Ca²⁺ + influx by interacting with P/Q-type channels. *Cell Rep* 2021; 35:109248. <https://doi.org/10.1016/j.celrep.2021.109248>
15. Valente P, Romei A, Fadda M, Sterlini B, Lonardoni D, Forte N, Fruscione F, Castroflorio E, Michetti C, Giansante G, Valtorta F, Tsai JW, Zara F, Nieuws T, Corradi A, Fassio A, Baldelli P, Benfenati F (2019) Constitutive inactivation of the PRRT2 gene alters short-term synaptic plasticity and promotes network hyperexcitability in hippocampal neurons. *Cereb Cortex* 29:2010–2033. <https://doi.org/10.1093/cercor/bhy079>
16. Fruscione F, Valente P, Sterlini B, Romei A, Baldassari S, Fadda M, Prestigio C, Giansante G, Sartorelli J, Rossi P, Rubio A, Gambardella A, Nieuws T, Broccoli V, Fassio A, Baldelli P, Corradi A, Zara F, Benfenati F (2018) PRRT2 controls neuronal excitability by negatively modulating Na⁺ + channel 1.2/1.6 activity. *Brain* 141:1000–1016. <https://doi.org/10.1093/brain/awy051>
17. Binda F, Valente P, Marte A, Baldelli P, Benfenati F (2021) Increased responsiveness at the cerebellar input stage in the PRRT2 knockout model of paroxysmal kinesigenic dyskinesia. *Neurobiol Dis* 152:105275. <https://doi.org/10.1016/j.nbd.2021.105275>
18. Sterlini B, Romei A, Parodi C, Aprile D, Oneto M, Aperia A, Valente P, Valtorta F, Fassio A, Baldelli P, Benfenati F, Corradi A (2021) An interaction between PRRT2 and Na⁺/K⁺ + ATPase contributes to the control of neuronal excitability. *Cell Death Dis* 12:292. <https://doi.org/10.1038/s41419-021-03569-z>
19. Tan GH, Liu YY, Wang L, Li K, Zhang ZQ, Li HF, Yang ZF, Li Y, Li D, Wu MY, Yu CL, Long JJ, Chen RC, Li LX, Yin LP, Liu JW, Cheng XW, Shen Q, Shu YS, Sakimura K, Liao LJ, Wu ZY, Xiong ZQ (2018) PRRT2 deficiency induces paroxysmal kinesigenic dyskinesia by regulating synaptic transmission in cerebellum. *Cell Res* 28:90–110. <https://doi.org/10.1038/cr.2017.128>
20. Mo J, Wang B, Zhu X, Wu X, Liu Y (2019) PRRT2 deficiency induces paroxysmal kinesigenic dyskinesia by influencing synaptic function in the primary motor cortex of rats. *Neurobiol Dis* 121:274–285. <https://doi.org/10.1016/j.nbd.2018.10.011>
21. Calame DJ, Xiao J, Khan MM, Hollingsworth TJ, Xue Y, Person AL, LeDoux MS (2020) Presynaptic PRRT2 deficiency causes cerebellar dysfunction and paroxysmal kinesigenic dyskinesia. *Neuroscience* 448:272–286. <https://doi.org/10.1016/j.neuroscience.2020.08.034>
22. Kuo CC, Chen RS, Lu L, Chen RC (1997) Carbamazepine inhibition of neuronal Na⁺ + currents: quantitative distinction from phenytoin and possible therapeutic implications. *Mol Pharmacol* 51:1077–1083. <https://doi.org/10.1124/mol.51.6.1077>
23. Dale RC, Gardiner A, Branson JA, Houlden H (2014) Benefit of carbamazepine in a patient with hemiplegic migraine associated with PRRT2 mutation. *Dev Med Child Neurol* 56:910. <https://doi.org/10.1111/dmcn.12396>

24. Mao CY, Shi CH, Song B, Wu J, Ji Y, Qin J, Li YS, Wang JJ, Shang DD, Sun SL, Xu YM (2014) Genotype-phenotype correlation in a cohort of paroxysmal kinesigenic dyskinesia cases. *J Neurol Sci* 340:91–93. <https://doi.org/10.1016/j.jns.2014.02.034>
25. Tran HT, Nguyen KV, Vercueil L (2021) Successful Treatment of a Paroxysmal Kinesigenic Dyskinesia Patient with carbamazepine-Induced Stevens-Johnson syndrome using oxcarbazepine monotherapy: a case report. *Case Rep Neurol* 13:598–604. <https://doi.org/10.1159/000518891>
26. Abriel H, Kass RS (2005) Regulation of the voltage-gated cardiac sodium channel Nav1.5 by interacting proteins. *Trends Cardiovasc Med* 15:35–40. <https://doi.org/10.1016/j.tcm.2005.01.001>
27. Yu FH, Westenbroek RE, Silos-Santiago I, McCormick KA, Lawson D, Ge P, Ferriera H, Lilly J, DiStefano PS, Catterall WA, Scheuer T, Curtis R (2003) Sodium channel beta4, a new disulfide-linked auxiliary subunit with similarity to beta2. *J Neurosci* 23:7577–7585. <https://doi.org/10.1523/JNEUROSCI.23-20-07577.2003>
28. Calhoun JD, Isom LL (2014) The role of non-pore-forming β subunits in physiology and pathophysiology of voltage-gated sodium channels. *Handb Exp Pharmacol* 221:51–89. https://doi.org/10.1007/978-3-642-41588-3_4
29. Laemmli UK (1970) Cleavage of structural proteins during the assembly of the head of bacteriophage T4. *Nature* 227:680–685. <https://doi.org/10.1038/227680a0>
30. Neher E (1992) Correction for liquid junction potentials in patch clamp experiments. *Meth Enzymol* 207:123–131. [https://doi.org/10.1016/0076-6879\(92\)07008-C](https://doi.org/10.1016/0076-6879(92)07008-C)
31. Bant JS, Raman IM (2010) Control of transient, resurgent, and persistent current by open-channel block by Na channel beta4 in cultured cerebellar granule neurons. *Proc Natl Acad Sci USA* 107:12357–12362. <https://doi.org/10.1073/pnas.1005633107>
32. Stafstrom CE, Schwindt PC, Crill WE (1984) Repetitive firing in layer V neurons from cat neocortex in vitro. *J Neurophysiol* 52:264–277. <https://doi.org/10.1152/jn.1984.52.2.264>
33. Khaliq ZM, Gouwens NW, Raman IM (2003) The contribution of resurgent sodium current to high-frequency firing in Purkinje neurons: an experimental and modeling study. *J Neurosci* 23:4899–4912. <https://doi.org/10.1523/JNEUROSCI.23-12-04899.2003>
34. Bean BP (2005) The molecular machinery of resurgent sodium current revealed. *Neuron* 45:185–187. <https://doi.org/10.1016/j.neuron.2005.01.006>
35. Aman TK, Grieco-Calub TM, Chen C, Rusconi R, Slat EA, Isom LL, Raman IM (2009) Regulation of persistent Na current by interactions between beta subunits of voltage-gated Na channels. *J Neurosci* 29:2027–2042. <https://doi.org/10.1523/JNEUROSCI.4531-08.2009>
36. Chen Y, Yu FH, Sharp EM, Beacham D, Scheuer T, Catterall WA (2008) Functional properties and differential neuromodulation of Na(v)1.6 channels. *Mol Cell Neurosci* 38:607–615. <https://doi.org/10.1016/j.mcn.2008.05.009>
37. Michetti C, Corradi A, Benfenati F (2017) PRRT2, a network stability gene. *Oncotarget* 8:55770–55771. doi: 10.18632/oncotarget.19506

38. Lerche H (2018) Synaptic or ion channel modifier? PRRT2 is a chameleon-like regulator of neuronal excitability. *Brain* 141(4):938–941. <https://doi.org/10.1093/brain/awy073>
39. Kruger LC, Isom LL (2016) Voltage-Gated Na⁺ Channels: Not Just for Conduction. *Cold Spring Harb Perspect Biol* 8:a029264. doi: 10.1101/cshperspect.a029264
40. Chen C, Calhoun JD, Zhang Y, Lopez-Santiago L, Zhou N, Davis TH, Salzer JL, Isom LL (2012) Identification of the cysteine residue responsible for disulfide linkage of Na⁺ channel α and β 2 subunits. *J Biol Chem* 287:39061–39069. <https://doi.org/10.1074/jbc.M112.397646>
41. Das S, Gilchrist J, Bosmans F, Van Petegem F (2016) Binary architecture of the Nav1.2- β 2 signaling complex. *Elife* 5:e10960. <https://doi.org/10.7554/eLife.10960.001>
42. Yan Z, Zhou Q, Wang L, Wu J, Zhao Y, Huang G, Peng W, Shen H, Lei J, Yan N (2017) Structure of the Na(v)1.4- β 1 complex from electric eel. *Cell* 170:470–482. <https://doi.org/10.1016/j.cell.2017.06.039>
43. Zimmer T, Benndorf K (2007) The intracellular domain of the β 2 subunit modulates the gating of cardiac Na⁺ v 1.5 channels. *Biophys J* 92:3885–3892. <https://doi.org/10.1529/biophysj.106.098889>
44. Grieco TM, Malhotra JD, Chen C, Isom LL, Raman IM (2005) Open-channel block by the cytoplasmic tail of sodium channel β 4 as a mechanism for resurgent sodium current. *Neuron* 45:233–244. <https://doi.org/10.1016/j.neuron.2004.12.035>
45. Schmidt J, Rossie S, Catterall WA (1985) A large intracellular pool of inactive Na channel α subunits in developing rat brain. *Proc Natl Acad Sci USA* 82:4847–4851. <https://doi.org/10.1073/pnas.82.14.4847>
46. Waechter CJ, Schmidt JW, Catterall WA (1983) Glycosylation is required for maintenance of functional sodium channels in neuroblastoma cells. *J Biol Chem* 258:5117–5123. [https://doi.org/10.1016/S0021-9258\(18\)32546-8](https://doi.org/10.1016/S0021-9258(18)32546-8)
47. Schmidt JW, Catterall WA (1987) Palmitoylation, sulfation, and glycosylation of the α subunit of the sodium channel. Role of post-translational modifications in channel assembly. *J Biol Chem* 262:13713–13723. [https://doi.org/10.1016/S0021-9258\(19\)76485-0](https://doi.org/10.1016/S0021-9258(19)76485-0)
48. O'Malley HA, Isom LL (2015) Sodium channel β subunits: emerging targets in channelopathies. *Annu Rev Physiol* 77:481–504. <https://doi.org/10.1146/annurev-physiol-021014-071846>
49. Molinarolo S, Lee S, Leisle L, Lueck JD, Granata D, Carnevale V, Ahern CA (2018) Cross-kingdom auxiliary subunit modulation of a voltage-gated sodium channel. *J Biol Chem* 293:4981–4992. <https://doi.org/10.1074/jbc.RA117.000852>
50. Laedermann CJ, Syam N, Pertin M, Decosterd I, Abriel H (2013) β 1- and β 3- voltage-gated sodium channel subunits modulate cell surface expression and glycosylation of Nav1.7 in HEK293 cells. *Front Cell Neurosci* 7:137. <https://doi.org/10.3389/fncel.2013.00137>
51. Corta da E, Brugada R, Verges M (2019) N-Glycosylation of the voltage-gated sodium channel β 2 subunit is required for efficient trafficking of Na(V)1.5/ β 2 to the plasma membrane. *J Biol Chem* 294:16123–16140. <https://doi.org/10.1074/jbc.RA119.007903>

52. Kazarinova-Noyes K, Malhotra JD, McEwen DP, Mattei LN, Berglund EO et al (2011) Contactin associates with Na⁺ channels and increases their functional expression. *J Neurosci* 21:7517–7525. <https://doi.org/10.1523/JNEUROSCI.21-19-07517.2001>
53. McEwen DP, Meadows LS, Chen C, Thyagarajan V, Isom LL (2014) Sodium channel β 1 subunit-mediated modulation of Nav1.2 currents and cell surface density is dependent on interactions with contactin and ankyrin. *J Biol Chem* 279:16044–16049. <https://doi.org/10.1074/jbc.M400856200>

Figures

Figure 1

PRRT2 and the Na_v β 4-subunit do not interact or compete for binding to Na_v1.2.

A. Left: Representative immunoblot of co-immunoprecipitation of β 1/ β 4-subunits by PRRT2. HA-tagged PRRT2 (PRRT2), bacterial alkaline phosphatase (BAP) and FLAG-tagged β 1/ β 4-subunits were expressed in wild type HEK293 cells. Cells lysates (INPUT, 10 μ g protein) and samples immunoprecipitated by anti-HA beads (HA-pellet) were analyzed by western blotting with anti-FLAG and anti-HA antibodies. The representative blots were cut from the same gel. **Right:** Quantification of the FLAG immunoreactive signal in PRRT2 immunoprecipitates normalized to the control BAP values. Means $\pm$ sem of $n = 3$ independent experiments. **B. Left:** Representative immunoblot of co-immunoprecipitation of PRRT2 and Na_v1.2 from extracts of Na_v1.2-expressing stable HEK293 clones transiently transfected with either HA-tagged PRRT2 alone or with HA-PRRT2 + β 4 subunit (β 4). BAP was used as a control. Cells lysates (INPUT, 10 μ g protein) and samples immunoprecipitated by anti-HA beads (HA-pellet) were analyzed by western blotting with anti-panNa_v and anti-HA antibodies. **Right:** Quantification of the Na_v immunoreactivity in PRRT2 immunoprecipitates expressed as ratios between normalized Na_v1.2 and PRRT2 immunoreactivities. Means $\pm$ sem of $n = 3$ independent experiments. **C. Left:** Representative immunoblot of co-immunoprecipitation of Na_v1.2 by PRRT2 in the presence or absence of an excess of β 4-subunit. HA-tagged PRRT2 or BAP was expressed in Na_v1.2-expressing HEK293 cells, whereas the β 4-subunit (β 4) was overexpressed in wild-type HEK293 cells. The extract from the β 4-subunit expressing cells was added to the HA-immunoprecipitated PRRT2/Nav 1.2 complex. Cells lysates (INPUT, 10 μ g protein) and samples immunoprecipitated by anti-HA beads (HA-pellet) were analyzed by western blotting with anti-panNa_v and anti-HA antibodies. **Right:** Quantification of the Na_v immunoreactivity in PRRT2 immunoprecipitates normalized to the BAP values. Means $\pm$ sem of $n = 3$ independent experiments.



Figure 2

PRRT2 and the Na_v β4-subunit have opposite effects on membrane targeting and exposure of Na_v1.2 channels.

A. Schematics of the biotinylation experiment. **B.** Representative immunoblots of cell surface biotinylation performed in HEK293 cells expressing Na_v1.2 and transfected with empty vector (MOCK),

PRRT2-HA, $\beta 4$ -FLAG or both. Total lysates (Input; *left*) and biotinylated (cell surface; *right*) fractions were analyzed by western blotting. Membranes were probed with antibodies to panNa_v, HA, FLAG and Na/K-ATPase (Na/K), with the latter used as marker of cell surface fractions. **C.** The cell surface Na_v immunoreactivity is expressed in percent of the control MOCK value after normalization to Na/K-ATPase immunoreactivity. Means $\pm$ sem of $n = 4$ independent experiments. Two-way ANOVA revealed no significant interaction between PRRT2 and $\beta 4$ -subunit ($F_{1,12} = 3.238$; $p=0.1$). * $p<0.05$, ** $p<0.01$; one-way ANOVA/Fischer's Least Significant Difference tests *versus* MOCK.

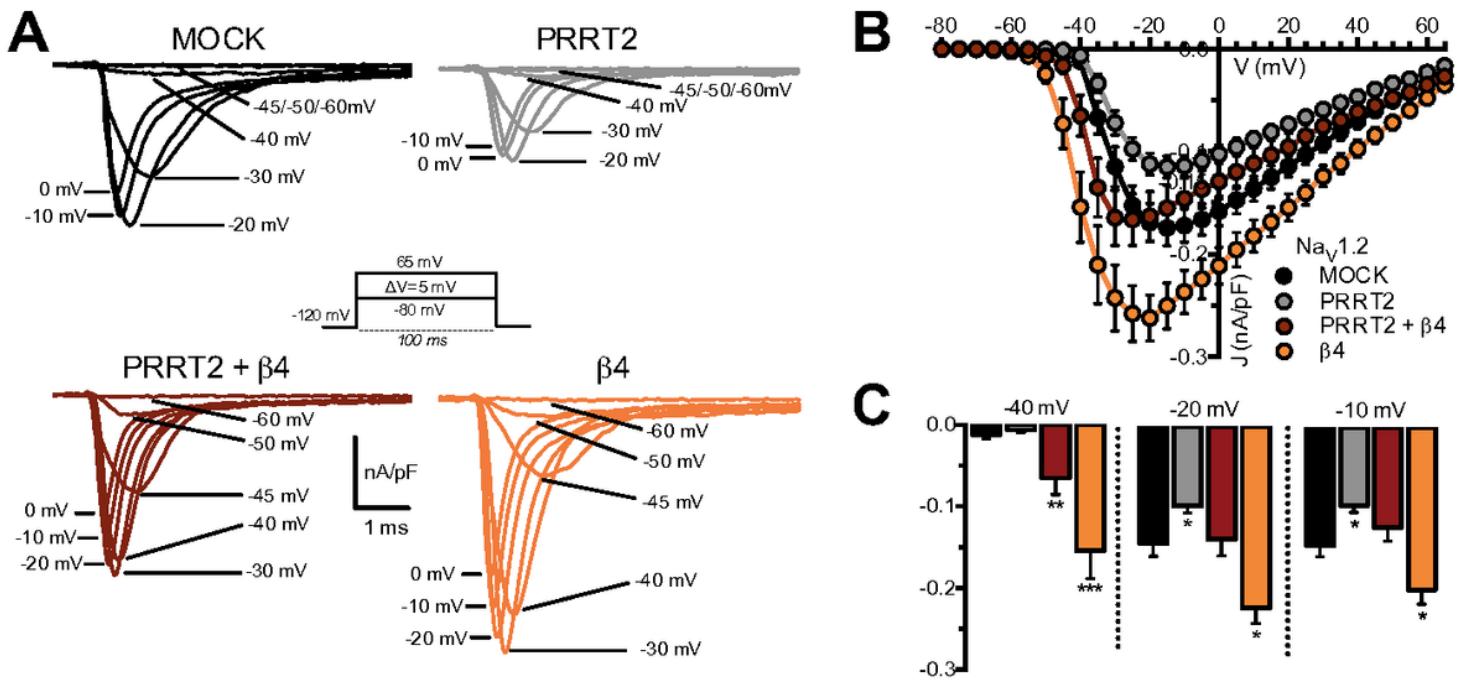


Figure 3

Figure 3

PRRT2 and the $\beta 4$ -subunit have opposite effects on the expression of the Na_v1.2 transient current.

A. Representative whole-cell transient Na⁺ currents recorded in HEK293 cells stably expressing Na_v1.2 and transiently transfected with empty vector (MOCK, black), PRRT2 (gray), $\beta 4$ -subunit (orange) or PRRT2+ $\beta 4$ subunit (dark red). Currents were elicited by a protocol (*inset*) consisting of 5-mV depolarization steps from -80 to 65 mV from a holding potential of -120 mV. For clarity, the first 6 ms of the 100-ms steps for eight representative traces per condition are plotted. **B,C.** Current density (J) versus voltage (V) relationship (**B**) for all the studied experimental conditions. The statistical analysis of J values at three representative voltages (-40/-20/-10 mV) is reported (**C**). Data are expressed as means $\pm$ sem (MOCK, $n = 36$; PRRT2, $n = 22$; $\beta 4$, $n = 18$; PRRT2+ $\beta 4$, $n = 17$). Two-way ANOVA revealed no significant interaction between PRRT2 and $\beta 4$ -subunit on the amplitude of the macroscopic Na⁺ current (-40 mV:

$F_{1,88} = 0.021$; $p=0.88$; -20 mV: $F_{1,90} = 1.2$; $p=0.276$; -10 mV: $F_{1,89} = 0.804$; $p=0.37$). * $p<0.05$, ** $p<0.01$, *** $p<0.001$ *versus* MOCK, Kruskal-Wallis/Dunn's tests.

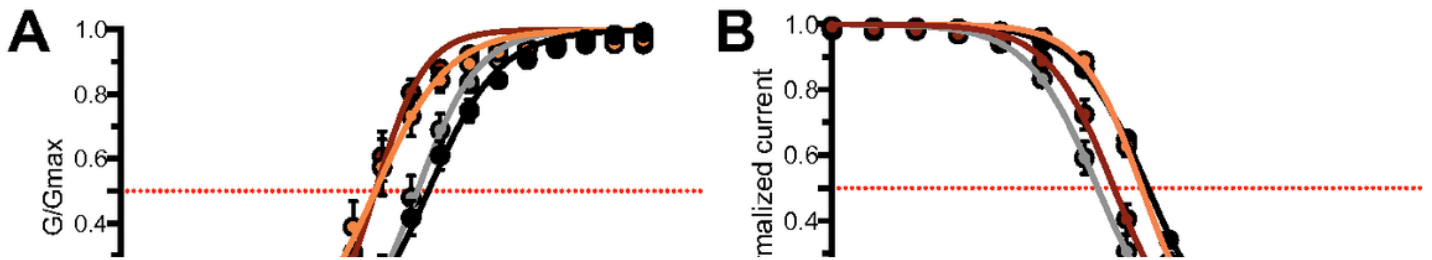


Figure 4

PRRT2 and β4-subunit differentially affect the kinetics of activation and inactivation of Na_v1.2 channels.

A,B. Voltage-dependence of activation (**A**) and steady-state inactivation (**B**) curves fit with a Boltzmann function for all conditions tested. Activation was studied using recordings obtained with the protocol depicted in Figure 3A. Steady-state inactivation was obtained with a protocol in which the cell under study was held at a series of voltages ranging from -130 mV to 30 mV for 500 ms followed by a 20 -ms step pulse to -10 mV to measure channel availability using a holding potential of -120 mV. **C,D.** Mean parameters of activation (**C**) and steady-state inactivation (**D**) curves fits to data obtained from all condition tested. All data are expressed as means $\pm$ sem. Activation: MOCK, $n = 36$; PRRT2, $n = 22$; β4, $n = 18$; PRRT2+β4, $n = 17$. Steady-state inactivation: MOCK, $n = 44$; PRRT2, $n = 16$; β4, $n = 28$; PRRT2+β4, $n =$

24. No significant differences were found in the activation kinetics between $\beta 4$ -subunit alone and PRRT2 + $\beta 4$ -subunit groups, as well as in the inactivation kinetics between PRRT2 alone and PRRT2 + $\beta 4$ -subunit groups. Two-way ANOVA revealed no significant interaction between PRRT2 and $\beta 4$ -subunit on the analyzed biophysical parameters (activation $V_{0.5}$: $F_{1,89} = 0.744$, $p=0.39$; G_{max} : $F_{1,89} = 0.645$, $p=0.424$; inactivation $V_{0.5}$: $F_{1,108} = 3.57$, $p=0.061$; inactivation slope: $F_{1,108} = 1.510$, $p=0.221$). * $p<0.05$, ** $p<0.01$, *** $p<0.001$ *versus* MOCK; one-way ANOVA/Bonferroni's tests or Kruskal-Wallis/Dunn's tests.

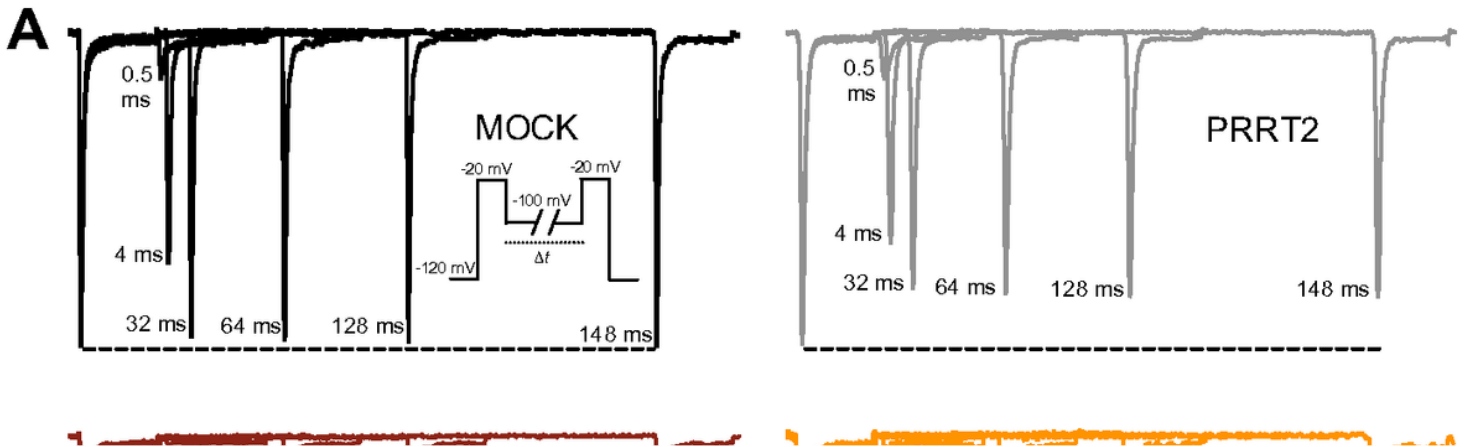


Figure 5

PRRT2, but not the $\beta 4$ -subunit, modulates the $\text{Na}_v1.2$ recovery kinetics from fast inactivation.

A. Representative channel recovery from inactivation current traces recorded for all the experimental conditions. Recordings were obtained pre-pulsing cells to -20 mV for 20 ms to inactivate Na^+ currents and then coming back to a recovery potential of -100mV for increasing durations before the repetition of test pulse to -20 mV. For clarity, 6 of the nine time intervals are shown. The dotted line represents the first pulse peak amplitude. **B.** Time courses of the recovery from inactivation of peak currents at -100 mV for all condition studied are plotted on a semi-logarithmic scale. **C.** Mean ($\pm$ sem) values of plateau and t of recovery estimated from one-phase decay fit to the data. **D.** Relationship between the 10th and the 1st test pulse evoked by protocol displayed in A. All data are expressed as means $\pm$ sem (MOCK, n = 46; PRRT2, n = 25; PRRT2 + $\beta 4$, n = 28; $\beta 4$, n = 29). Two-way ANOVA reveled no significant differences were found between PRRT2 alone and PRRT2 + $\beta 4$ -subunit groups (plateau of recovery: $F_{1,124} = 3.286$, $p = 0.072$; t of recovery: $F_{1,124} = 0.09$, $p = 0.764$; 10th/1st response: $F_{1,124} = 1.86$, $p = 0.174$). ** $p < 0.01$, *** $p < 0.001$ *versus* MOCK; Kruskal-Wallis/Dunn's tests.

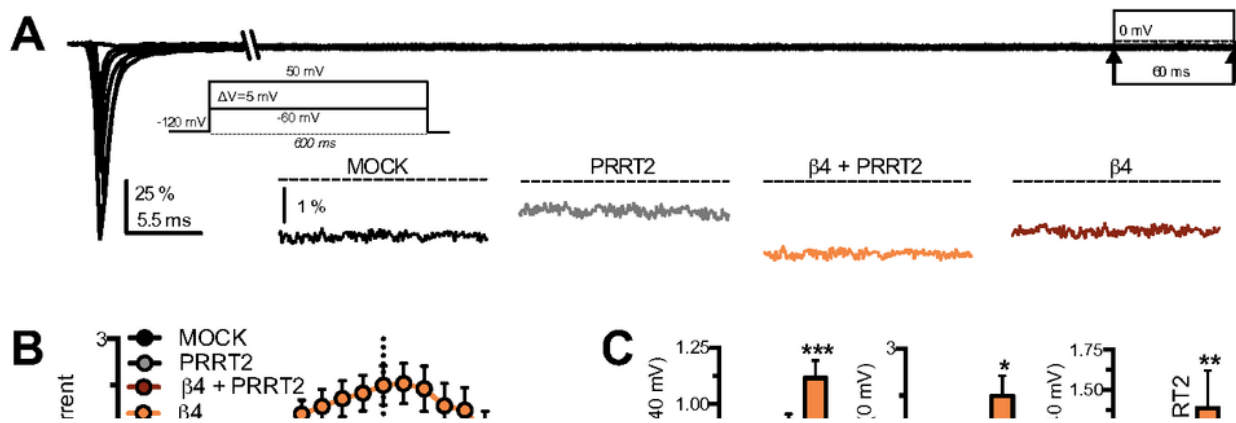


Figure 6

PPRT2 and β4-subunit independently modulate the Na_v1.2 persistent and resurgent currents

A. Representative MOCK persistent Na_v1.2 currents evoked by depolarizing steps from -60 to 50 mV with 5-mV increments, lasting 600 ms (*inset*) in MOCK-transfected Na_v1.2-expressing HEK293. For clarity, only currents evoked at -60, -40, -20, 0, 20, and 40 mV are reported. The highlighted box at the end of

stimulation indicates the region of the trace in which the persistent current was measured. The insets show zoomed views of the maximal persistent current for all conditions tested. **B.** Persistent current, measured as the mean current in the last 60 ms of each 600 ms step and normalized to the transient current peak, is plotted *versus* voltage in each cell. **C.** Bar plots of the mean ($\pm$ sem) values of the normalized persistent current amplitude recorded at three distinct voltages (-40, 0 and 40 mV) for all tested conditions. Data are expressed as means $\pm$ sem (MOCK, n = 24; PRRT2, n = 10; PRRT2+ β 4, n = 20; β 4, n = 21). **D.** Representative peak resurgent current traces generated by Na_v1.2-expressing HEK293 cells either mock-transfected or transfected with PRRT2 recorded in the presence (yellow/red) or absence (black/grey) of the β 4 COOH-terminal peptide (β 4 ptd) in the intracellular recording solution. Currents were evoked with a family of steps depolarizations from -120 mV to 30 mV for 20 ms to open the channels, allow them to undergo open-channel block and subsequently repolarize to a different potential ranging from -50 mV to 20 mV for 60 ms to allow the blocker to unbind. For clarity, only the peak resurgent current trace for all condition is plotted. **E.** Resurgent currents evoked during repolarization were normalized to peak transient current at each voltage for all experimental conditions. Peak resurgent current was measured for each voltage after 2.5 ms into the repolarization step to bypass fast tail currents. **F.** Bar plots of the mean ($\pm$ sem) values of the normalized resurgent current amplitude recorded at three distinct voltages at -20 and -50 mV for all conditions tested. All data are expressed as means $\pm$ sem (MOCK, n = 6; MOCK+ β 4 ptd, n = 12; PRRT2, n = 6; PRRT2+ β 4 ptd, n = 14). Two-way ANOVA revealed no significant interaction between PRRT2 and β 4-subunit on the amplitude of both the persistent and the resurgent Na⁺ currents (Persistent at: -40 mV, $F_{1,71} = 0.002$, $p=0.988$; 0 mV: $F_{1,71} = 0.00519$, $p=0.942$; 40 mV, $F_{1,71} = 1.94$, $p=0.167$. Resurgent at: -50 mV, $F_{1,34} = 0.047$, $p=0.828$; -20 mV: $F_{1,34} = 0.00043$, $p=0.983$). * $p<0.05$, ** $p<0.01$, *** $p<0.001$, *versus* MOCK, one-way ANOVA/Bonferroni or Kruskal-Wallis/Dunn tests.

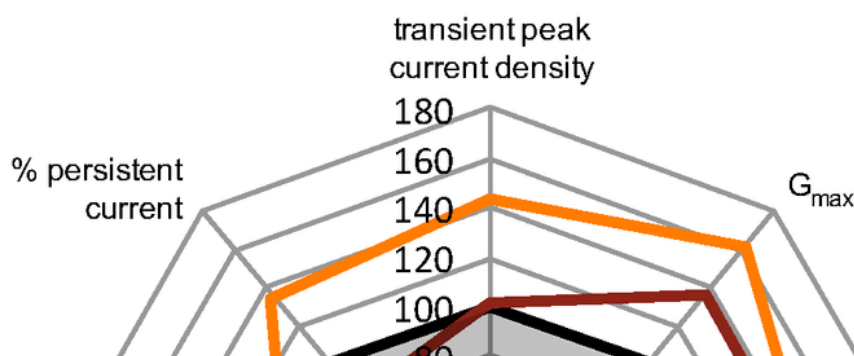


Figure 7

Summary of the differential modulation of Na_v1.2 channels by PRRT2 and the β4-subunit.

Radar plot showing the effects of the single or combined expression of PRRT2 and Na_v β4-subunit on Na_v1.2 properties. The plot was built with each spoke arranged in such a way that points lying outside the control condition (MOCK plot; black) are indicative of a gain-of-function, while points lying within the

MOCK plot indicate a loss-of-function. All parameters are expressed in percent of the MOCK group. Values of transient peak current density were recorded at -20 mV. The percentage values of persistent current were obtained from recordings performed at 0 mV.